

Muhammad Farhan Farooq

Cell# +49-16097580950

Email ID: farhan.farooq@gmail.com



Senior Telecommunications Professional:

Dynamic and results-oriented telecom expert with a proven track record of enhancing network performance, optimizing operations, and delivering customer-focused solutions. Adept at thriving in fast-paced environments, managing complex projects, and fostering strategic relationships to achieve business objectives. Recognized for technical expertise, leadership, and commitment to excellence in telecommunications.

Areas of Expertise:

Strategic Network Planning | Solution Consultancy | RAN Planning & Optimization | VoLTE/LTE/LTE-A/3G/2G | Access Management | Project Management | Customer Relationship Management | Critical Decision-Making

Vendor Experience: Huawei, Ericsson, ZTE, Siemens.

Work Experience - Details:

RAN Planning Consultant | February 2023 – till date

ORBIT IT-Solutions | Project: Deutsche Telekom – Germany

- Conducted meticulous network planning adhering to Deutsche Telekom's stringent guidelines to ensure optimal performance.
- Performed detailed site verifications, including antenna configurations, cell parameters, and system elements to maintain consistency with provider standards.
- Evaluated site planning documents to verify compliance with vendor protocols (Huawei, Ericsson).
- Designed coverage-focused site strategies to meet regional service objectives.
- Delivered actionable insights and recommendations during planning reviews, enabling seamless execution and enhanced service quality.

RF Team Lead | November 2020 – January 2023

Professional Employers Private Limited, ZTE Pakistan | Project: CMPAK – Pakistan

- Led the successful launch of USF sites in the South Region, ensuring adherence to tight deadlines and operational benchmarks.
- Strategically planned 2G/3G/4G network expansions to optimize coverage, throughput, and capacity.
- Coordinated weekly and monthly meetings with customers, presenting progress updates and addressing technical challenges.
- Spearheaded frequency band planning for L900/L1800/L2100/U2100, aligning with current network usage and optimizing spectral efficiency.
- Oversaw KPI design and acceptance testing, ensuring seamless site integrations.
- Directed initial tuning and optimization of on-air sites, achieving high performance in key acceptance metrics.
- Implemented advanced optimization techniques to enhance cluster performance and user experience.

Free Lance Consultant | August 2019 – July 2020

Conduct online courses for 5G-NR on following topics

- NR Use Cases, eMBB, mMTC, uRLLC. 5G-NR Frame structure & Channel Management
- Massive MIMO & Beamforming concept
- Concept of 5G Link Budget. PCI Planning Concept. PRAH, Root Sequence Planning Concept
- SCS (Subcarrier Spacing) wrt service types, frequency bands & moving speed
- Numerology concept wrt Time-domain/Frequency-domain and Space-domain resources.
- Massive MIMO concept, Beamforming concept
- 5G Uplink & DL Power Control
- 5G Scheduling, DL Scheduling/ UL Scheduling
- Mobility Management, NSA SgNB

- 5G Intraband CA
- Concept of 5G Link Budget with introduction of Foliage Loss/Body Block Loss & Rain Attenuation Margin
- 5G Planning Parameter
- Scenario-based MM Beam Configuration, Azimuth Planning/Tilt Planning
- PCI Planning Concept

LTE Team Lead | May 2017 – May 2019

LCC for Huawei (MS Project) | Project: CMPAK – Pakistan

- Worked on FDD network with 10M /15M/20 & Compact Band 18.6 MHz
- Successfully managed the launch of VoLTE and SRVCC in CMPAK's network within strict timelines, collaborating closely with Core, IMS, and BSS teams.
- Conducted VOLTE features trial, other than basic VOLTE, TTI Bundling/Voice Characteristics Awareness Scheduling/ RObust Header Compression (ROHC) for Capacity Enhancement & Coverage Improvement.
- Optimization of VoLTE and Violate
 - ✓ PDCP/RLC Optimization
 - ✓ VoLTE Parameter – Timer
 - ✓ VoLTE Feature – VQI (VQM to monitor user voice quality)
 - ✓ VoLTE- Inter-Freq /Inter-FreqHO
- Responsible for CA implementation 2CC & 3CC for DL and 2CC for UL
- MLB for Load Balance on High Congested Sites
- Optimization of GUL Interworking parameters to ensure that selection, re-selection, handover, redirection, CSFB and other RAT priority strategies are properly optimized to ensure efficient interworking.
- Execution of multiple Trials to improve various KPIs like UL/DL Throughput/HHOSR/CDR/CSSR/Latency etc.
- Responsible for Benchmarking Project.
- Involve in Daily Dashboard template making and finalization
- Responsible for Bandwidth upgradation/expansion activity from 10M to 15M & 15M to 20M
- Responsible to interact with Customer for Weekly/Monthly meetings
- Coordinated multi-departmental operations to streamline processes and align project goals with client requirements.

3G/4G RF PLANNING & OPTIMIZATION Team Lead | February 2015 – February 2017

Vintur for Huawei (MS Project) | Project: MTN – South Africa

- Led benchmarking projects to secure network performance leadership, achieving superior P3 (now umlaut) scores.
- Directed FDD network planning and implementation for 3M/5M/10M bands, ensuring capacity and coverage goals.
- Conducted capacity audits and proposed expansion strategies to support growing subscriber demands.
- Planned and implemented multi-carrier strategies to enhance throughput and user experience.

3G/4G RF PLANNING & OPTIMIZATION Advisor | June 2014 – December 2014

HRSG PAKISTAN | Project: Telenor – Pakistan

- Responsible for successful 3G deployment.
- Main responsibilities are to manage vendor (ZTE), RAN algorithm development and smooth execution with 2G algorithm.
- Responsible to train whole RAN organization on the formulation of optimization principles
- Responsible to provide all integration, configurations, testing, Data fills

3G/4G RF PLANNING & OPTIMIZATION LEAD | December 2012 – May 2014

Alkan East L.L.C OMAN for Huawei (Rollout Project) | Project: Omantel – Oman

- Coordinating with Project and Marketing team for RF related issues. Regular customer meeting & taking care of the requirements & queries.
- LTE rollout of commercial network, including planning (PCI, PRACH, TA), Single Site verification of LTE Sites.
- LTE Cluster Optimization (Improve RSRP/SINR/Accessibility/Retainability/CSFB).
- Successfully done trial of MIMO 4T4R
- Responsible for implementation of HSPA+ in the network. Finalized strategy with solution team.
- Successfully Implemented UMTS 2nd Carrier with U2100 in the network
- Design and implemented strategy for 3rd carrier with U900 for Coverage
- Responsible for Network Antenna Modernization Project by swapping single band to multi band Antenna

RF PLANNING & OPTIMIZATION Lead | May 2011 – May 2012

Vintur for Huawei (Rollout Project) | Project: Vodacom – South Africa

- Responsible for New Site Planning & Script Preparation and execution
- Responsible for New Site Planning for coverage/capacity & PSC /PRACH Planning
- VIP call complaints analysis & recommendations for the improvement.

3G RF PLANNING & OPTIMIZATION ENGINEER | March 2010 – May 2011

Advacom for Huawei (MS Project) | Project: Etisalat - UAE

- Monitor and analyze KPI Network statistics on daily basis.
- PS/HSDPA/HSPA+ performance improvement (Accessibility, Retainability and throughput).
- VIP Complain analysis and resolution using Call Trace

3G RF PLANNING & OPTIMIZATION ENGINEER | December 2009 – March 2010

FCCI for Huawei (Rollout Project) | Project: VIVA - Bahrain

- Initial Tuning of Newly on Aired Sites along with script creation and its execution
- Cluster Optimization to meet all KPIs for Cluster acceptance agreed with the customer.
- 3G sites Neighbor planning according to Planning strategy agreed with the customer

RF Planning & Optimization Engineer | April 2008 – September 2009

AIRCOM (Rollout Project) | Project: Zain - Ghana

- Nominal Radio Network Planning using Planet for 2G/ 3G network.
- Performed 2G/3G Parameter Settings & Tuning for assigned Clusters
- Coverage analysis and recommendation to improve cluster RSCP and Ec/Io.
- Responsible for Cell planning, site surveys, site candidate selections, Scrambling Code Planning/Cell Data Parameters, Neighbor Relations Optimization, integration and acceptance of new sites

Assistant Manager RF | April 2005 – April 2008

Telenor Pakistan

- Managing team of 18 RF Engineers, including Telenor and LCC Planning Engineers
- Coordinating with Project/Implementation and Marketing team for RF related issues. Regular vendor meeting & taking care of the requirements & queries.
- Develop and execute planning & optimization team strategies to meet KPIs and ensure proper operations of access networks.
- Performing RF surveys and then preparing appropriate design and proposal products such as antenna location and schematic drawings.

Qualification:**B.E. (Electrical)**

N.E.D University of Engineering & Technology Karachi, Pakistan

TOOLS:

- MFiol/Winfiol for Command handling
- Ericsson OSS RNO/CNA
- Ericsson Business Object (BO) for Stats Monitoring
- ZTE CNO (Planning tool); Netmax (MR Analysis); U31 NetNumen for stats and configuration
- Aircom Asset
- ATOLL for prediction
- GENEX NASTAR
- ACTIX and TEMS Discovery
- Drive Test Tools: TEMS Investigation (Ericsson); GENEX PROBE.
- PLANET EV Planning Tool for Cell planning.
- GENEX ASSISTANT for post processing tool.
- Optima for stats monitoring
- Mapping Tool: MapInfo Professional
- Using Google Earth to perceive necessary information about the clutter
- MCOM/PIANO and AFP PLANET for Frequency Planning & MCOM/NASTAR tool for Neighbor Planning